Disruptions to Homeostasis

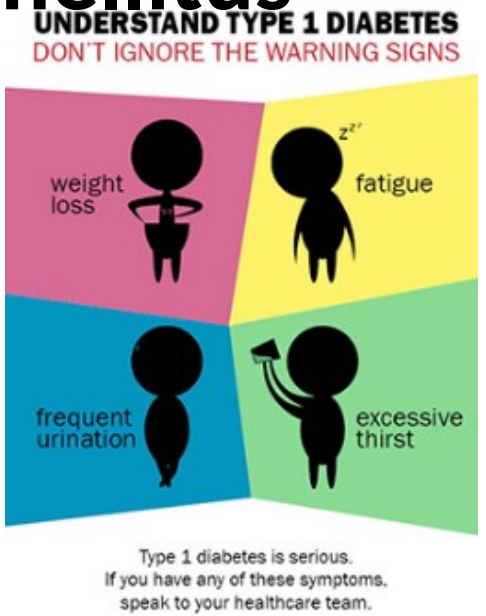


Review from several different chapters of *HP*

Disruptions to homeostasis

- Can cause serious disease and/or death if untreated
- Cells can only function within a narrow range of conditions.
- Disruptions focused on in Human Biology
  - Hormonal disruption
    - Diabetes Mellitus Types 1 and 2
    - Thyroid imbalance and disease
    - Growth hormone imbalance and disease
  - Behavioural causes of disruption
    - Lifestyle choices – exercise, food, choices to use drugs, alcohol
  - Diseases of body systems
    - Emphysema
    - High blood pressure
    - Injury

Hormonal Causes of Disruption – Diabetes mellitus



- Diabetes Mellitus
  - Means “sweet urine”
  - Two types – Type 1 and Type 2
- Type 1 Diabetes:
  - Also called “insulin-dependent” diabetes
  - Childhood onset
  - Immune destruction of beta cells in Islets of Langerhans
  - No insulin produced
  - Glucose can’t enter cells – body starts to break down protein → ketoacidosis
  - Symptoms:
    - Tiredness - cells being unable to access glucose for metabolism
    - Extreme thirst - glucose in blood raises osmotic pressure – how does this cause thirst?
    - Frequent urination - increased drinking from thirst
    - Weight loss – cells cannot access glucose for metabolism
  - Treatment:
    - Insulin injections as it is not being made by the pancreas
    - Careful diet and exercise.

Hormonal Causes of Disruption – Diabetes mellitus

- Type 2 Diabetes: a lifestyle disease
  - Also called “adult-onset diabetes”
  - Can produce insulin, but cells do not respond by allowing glucose to enter.
  - Lifestyle factors involved:
    - Lack of physical activity
    - Being overweight or obese
    - High sugar, fat and salt diet, low fibre intake
    - High blood pressure
    - High cholesterol
    - Smoking
  - Symptoms:
    - Develops gradually
    - Extreme thirst – increased blood glucose increases osmotic pressure
    - Frequent urination – due to increased thirst
    - Weight loss (not always) – glucose can’t enter cells for metabolism.
    - Tiredness – glucose can’t enter cells for metabolism
  - Treatment:
    - better diet, exercise, lifestyle can reverse or control symptoms
    - Sometimes medication



Diabetes mellitus: a disorder of glucose regulation

Common Cause(s)	Symptoms	Diagnosis	Treatment
Type 1 Diabetes: • Autoimmune attack of pancreas destroys islet tissue. Little to no insulin produced	• Usually childhood onset • Extreme thirst • Excessive urination • Weakness • Fatigue • Diabetic ketoacidosis – odd smelling breath due to ketones • Death if untreated	• Blood test to show high blood glucose levels. • Presence of glucose in urine.	Daily insulin injections, accompanied by careful diet and exercise regime. 
Type 2 Diabetes: • Lifestyle cause with some genetic components. • Cells become resistant to insulin, so it cannot be absorbed	• Adult onset (45+) • Few symptoms at first • As severity increases: • Thirst • Excessive urination • Weakness • Poor wound healing • Headaches • Dizziness		• Lifestyle changes • Weight loss • Regular exercise • Low sugar, good quality diet • Medication to manage insulin resistance

Hormonal Causes of Disruption: Thyroid hormone imbalance and disease

- Problems with Thyroid hormone production affect homeostasis, causing disease.
- ***Hyperthyroidism*** – Too much Thyroid hormone produced
- ***Hypothyroidism*** – Too little Thyroid hormone produced
  - Lack of iodine required to make Thyroxine
- Over or under secretion of thyroid hormone affects metabolism (basal metabolic rate)
- Can be due to
  - Thyroid gland dysfunction
  - TSH secretion problems from the pituitary



Review: Normal Regulation of Thyroxine production

Hypothalamus

Detects:

- Low levels of Thyroid hormone in blood
- Low metabolic rate
  (including low body temperature, low blood sugar and other parameters)

The hypothalamus produces **Thryotropin Releasing Factor (TRF)** *which travels through the blood vessels of the infundibulum to the Anterior Pituitary*

Anterior Pituitary

The TRF stimulates the glandular cells of the Anterior Pituitary to produce **Thyroid Stimulating Hormone (TSH)**

TSH *is released from the Anterior Pituitary and travels through the systemic circulation*

Thyroid

The TSH binds to receptors in the Thyroid, stimulating it to produce **Thyroxine**

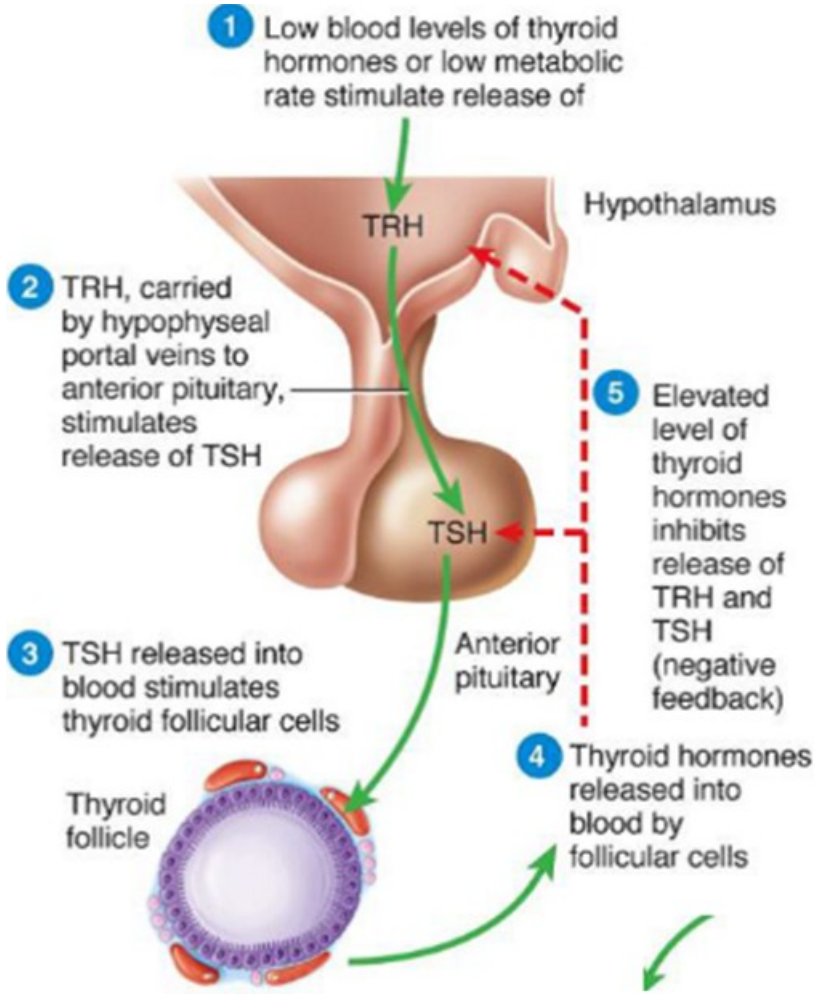
Thyroxine *is released from the Thyroid and travels through the systemic circulation*

Body Tissues

Thyroxine binds to receptors within most body tissues, which then respond by:

- Increasing metabolic rate and body temperature by increasing catabolism of protein and carbohydrates.
- Heart rate, blood pressure and respiratory rate are increased.
- Increases blood sugar by stimulating liver to convert glycogen to glucose and by stimulating intestines to absorb sugar.

Rising levels of Thyroxine inhibit further production of TRF and TSH, so that Thyroxine levels stabilise. This is **Negative Feedback**



Effects in Body

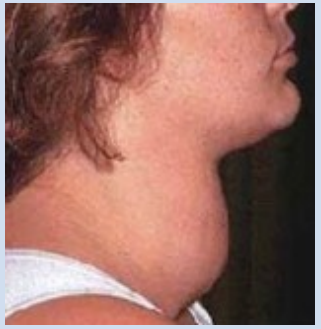
Increased Body Temperature
Increased BMR
Increased Blood Glucose
etc.

Hyperthyroidism

- Enlargement of thyroid and increased production of hormone due to immune system reaction
- “Graves’ Disease”
- Too much thyroxine produced – metabolism is boosted beyond homeostatic requirements:
  - Rapid heartbeat (to supply additional O_2 to cells and remove $CO_{2)}$
  - Weight loss (nutrients eg glucose are being metabolised more quickly)
  - Increased appetite (to bring in additional glucose for heightened metabolism)
  - Fatigue (difficulty in supplying cells with glucose etc for heightened metabolic need)
  - Sweating (increased body heat due to increased metabolism)
  - Anxiety
  - Sometimes protruding eyeballs
- Treatment:
  - Drugs that block use of iodine
  - Surgery to remove part or all of gland
  - Radioactive iodine therapy



Hyperthyroidism – too much thyroid hormone

Common causes:	Symptoms	Diagnosis	Treatment
Graves' Disease – immune system produces antibodies which stimulate thyroid to produce hormones **Thyroid nodules/adenoma** – iodine deficiency causes overgrowth of thyroid tissue to try to produce more hormone, which also impacts on TSH production. **Thyroiditis** – inflammation of the thyroid causes overproduction of hormone	Weight loss - usually (or gain sometimes) Increased appetite Protruding eyeballs Fast heart rate Sleep disturbances Anxiety/nervousness Fatigue Muscle weakness Goitre – from inflammation causing swelling of thyroid 	Blood test for TSH – lower than normal as excess thyroid hormone inhibits production of TSH via negative feedback. Testing for antibodies Radioactive iodine uptake by thyroid Ultrasound/CT scan/MRI 	Radioactive iodine therapy • given as medication • taken up by thyroid destroys thyroid cells • Less thyroid hormone produced Anti-thyroid medication • Prevents thyroid from using iodine to make hormones Surgery • Removes part or all of thyroid to lower production of thyroxine.

Hypothyroidism – too little thyroxine

- Too little thyroxine produced – so metabolism slowed below homeostatic requirements.
- More common than hyperthyroidism.
- Can be caused by problems with.
  - Thyroid gland producing too little TSH
  - Lack of iodine required to make Thyroxine
  - Pituitary production of TSH leading to under-stimulation of thyroxine
  - Hypothalamus producing to little TRF
- Symptoms:
  - Slow heart rate – metabolic rate is slower, not as much need for glucose and O_2 to get into cells.
  - Unexplained weight gain – cells are not metabolising glucose as fast for ATP, so glucose stored as fat.
  - Fatigue – slower metabolism means not enough ATP supplying muscles
  - Intolerance to cold – slower metabolism means less body heat produced.
  - Goitre – in cases where lack of iodine is the cause, the thyroid overgrows in an attempt to produce thyroxine, but can’t because there isn’t enough iodine to make the thyroxine.

Hypothyroidism – too little thyroid hormone

Common Causes	Symptoms	Diagnosis	Treatment
Hashimoto's Disease – autoimmune disease attacks thyroid tissue, destroying it so it cannot produce hormone. **Thyroid destroyed** by medication or surgery to correct hyperthyroidism **Not enough dietary iodine** – mostly a problem in developing nations – body can't produce hormone without iodine	Fatigue Weakness Weight gain Depression Cold intolerance Dry, rough skin Brittle dry hair Goitre 	Blood test – low levels of T4, high levels of TSH, as low T4 stimulates hypothalamus and pituitary to try to produce more hormone. If it is caused by a pituitary or hypothalamus issue so no TRF ors TSH is produced, the levels will be low, but this is far less common.	Thyroid hormone replacement medication: • *Levothyroxine* • Synthetic thyroid hormone tablets • Replace missing hormone 

Hormonal Causes of Disruption: Growth Hormone Imbalance and Disease

- Problems with Growth Hormone production affect homeostasis, causing disease.

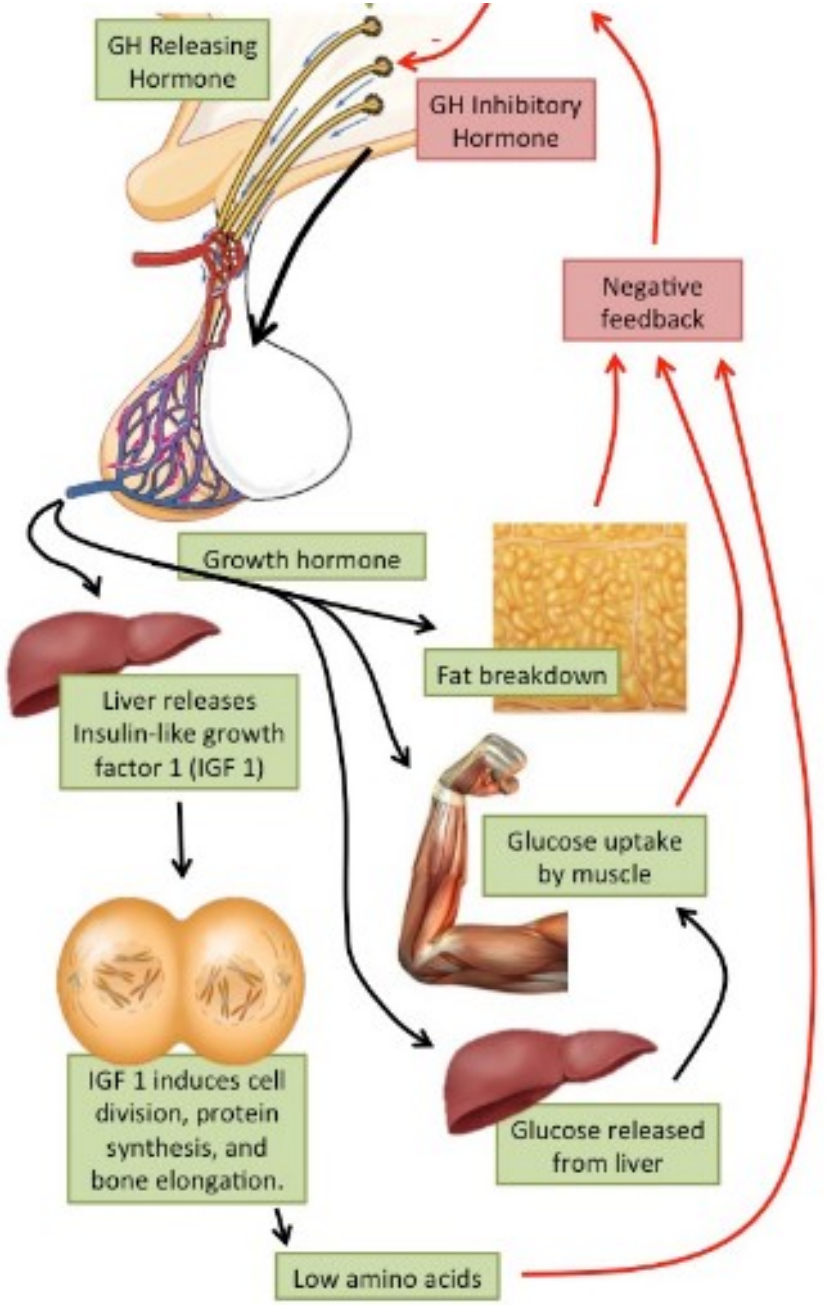
- ***Growth Hormone Hypersecretion***: too much growth hormone.

- ***Growth Hormone Deficiency***: not enough growth hormone.

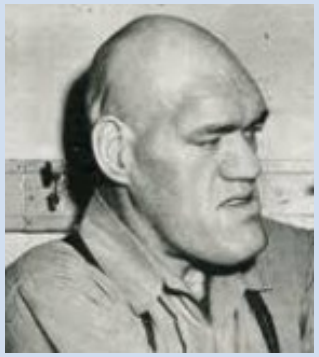


Review: Normal Growth Hormone Regulation

1. Growth Hormone Releasing Hormone (GHRF) released from Hypothalamus in response to:
   - Sex hormone interactions, eg during puberty
   - Action of certain neurotransmitters
   - Sleep (especially deep sleep)
   - Nutrition
   - Exercise
2. GHRF triggers Growth Hormone release from the anterior pituitary. Growth hormone enters the blood.
3. Growth hormone has effects on body as discussed.
4. Negative feedback:
   - Rising levels of Growth Hormone in blood, rising levels of fatty acids, glucose uptake by muscle, and low amino acid levels trigger release of **somatostatin (Growth Hormone Inhibitory Hormone/Factor)** from the Hypothalamus, which inhibits further Growth Hormone release by pituitary.



Growth Hormone Hypersecretion

Common Cause(s)	Symptoms	Diagnosis	Treatment
Commonly: Benign tumours on anterior pituitary gland causing oversecretion of Growth Hormone. Occasionally: as a result of inherited diseases or chromosomal abnormalities	Adults: *Acromegaly* • Larger than proportion hands and feet • Thick toes and fingers • Prominent jaw and forehead • Coarse facial features • Sweating • Headaches • Weakness/fatigue • Insomnia  Children: *Gigantism* • As above, but greater than normal height also.	Blood test for Insulin-Like Growth Factor Glucose tolerance test MRI to confirm presence of pituitary tumour(s) 	• Targeted radiation therapy to shrink tumours • Microsurgery to remove tumours • Somatostatin medication to inhibit GH release

Growth Hormone Deficiency

Common causes	Symptoms	Diagnosis	Treatment
Often no cause can be established in children Can go together with other pituitary diseases Childhood onset may be due to: • Heritable gene mutations • Congenital disease or malformation Adult onset may be due to pituitary damage from • Radiation • Surgery • Autoimmune attack of pituitary	Childhood onset: • Short stature – may be proportional or non-proportional. • Slow growth rate. • Slower muscle development. • Unusual facial features – prominent forehead. Adult onset: • Decrease in muscle and bone mass • Heart problems • Baldness (men) • Decreased energy and strength	Blood test to look for abnormally low growth hormone levels. 	Treatment with injections of growth hormone. • Recombinant gene technology • Gene to produce growth hormone spliced into bacteria • Genetically engineered growth hormone produced by bacteria

Behavioural Causes of Disruption

- Individual behaviours can be unhealthy and therefore disrupt homeostasis.
- For example:
  - Drugs: medicinal drugs assist with maintaining homeostasis, but non-prescription drugs eg. alcohol may disrupt homeostasis, often by interfering with neurotransmitters.
  - Excessive activity/exercise: exercise is good for you, but excessive exercise can put undue strain on the body eg tendons and joints.
  - Eating habits: these can contribute to lifestyle diseases, but can also contribute to malnutrition if intake is inadequate.

Disease Disrupts Homeostasis

- Diseases disrupt homeostasis by interfering with functioning of the body. Some examples include:
  - **Emphysema** – break down of alveoli – impacts on ability to maintain homeostatic levels of O_2 and CO_2.
  - **Very High Fever** – elevation in body temperature can impact ability to maintain chemical reactions required for life.
  - **High blood pressure** – inability to bring blood pressure into homeostatic range can interfere with kidney function and also increases risk of heart attack and stroke.
  - **Injury** – may cause blood loss, or allow infection entry, interfering with homeostasis.

Hormones and Recombinant DNA Technology - Review

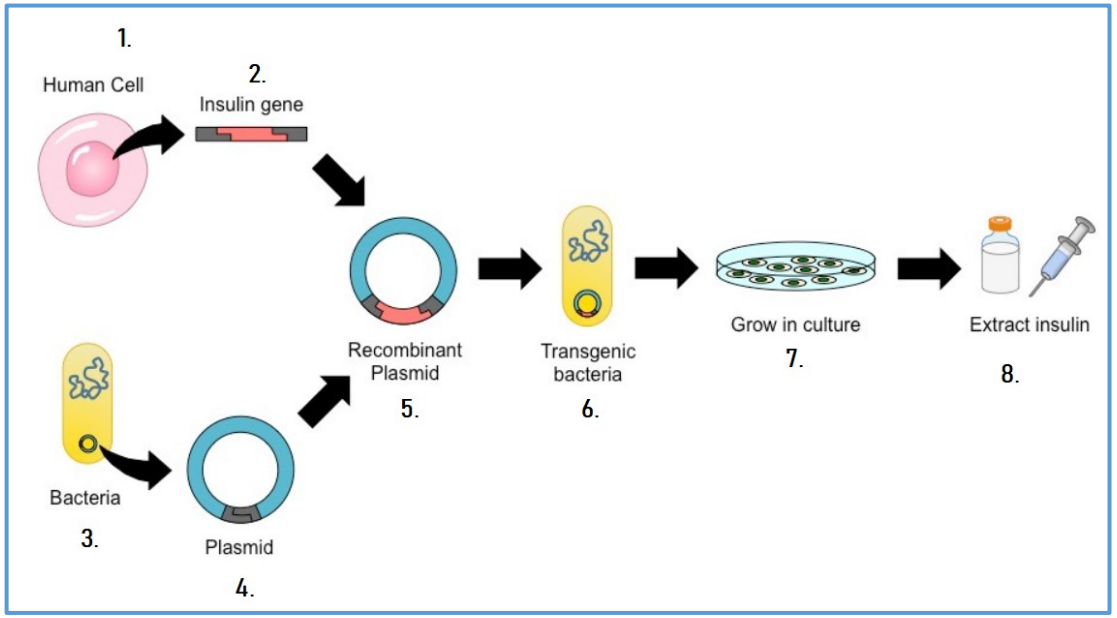


Recombinant DNA Technology

- Process of splicing DNA from one organism into another organism
- Variety of uses.
- Endocrine System uses of recombinant DNA Technology:
  - Synthetic Insulin
  - Synthetic Growth Hormone

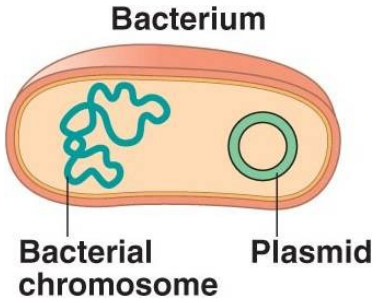
Putting a gene from one organism into another organism

- Technology is available to put a gene from one organism into a totally different organism!
- For example, scientists can put the insulin gene from a human cell into a bacterium, so that the bacterium produces the hormone insulin, which can then be extracted, refined and used to treat diabetes mellitus.
- This involves:
  - ✓ Isolating (cutting out) the insulin gene from a healthy human cell, using a *restriction enzyme*.
  - ✓ Treating a piece of bacterial DNA with the same *restriction enzyme*.
  - ✓ Splicing the human insulin gene into the bacterial DNA.
  - ✓ Sticking the spliced pieces together with the enzyme *DNA Ligase*.
  - ✓ Putting the bacterial DNA back into the bacterium.



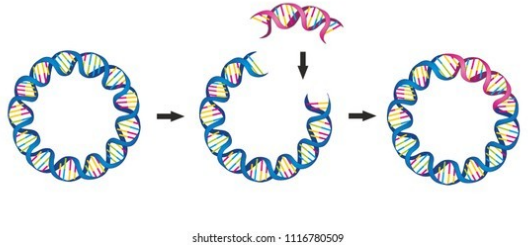
Splicing human genes into bacteria

- Bacteria have long strands of DNA, but also have smaller, circular strands of DNA in their cytoplasm.
- These small, circular strands of DNA are called *plasmids*.

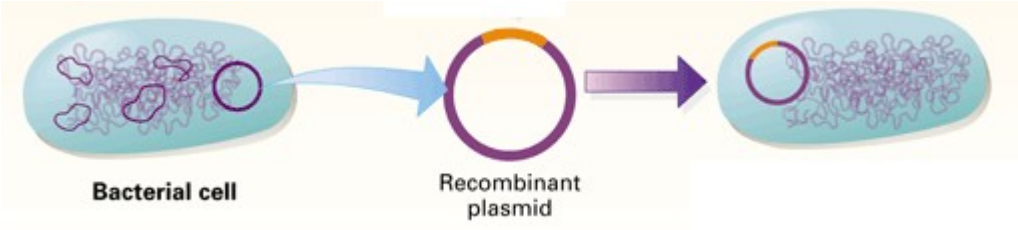


- *Plasmids* are removed from bacteria and treated with the same restriction enzymes used to isolate human genes, so the human gene can be spliced into the plasmid.

- The plasmid (now containing a human gene, eg for insulin) is a *vector* (carrier) that transports the human gene.



- The plasmid is placed into a new bacterium, so that bacterium will carry the human gene. If it is carrying the human insulin gene, it will produce the hormone insulin!



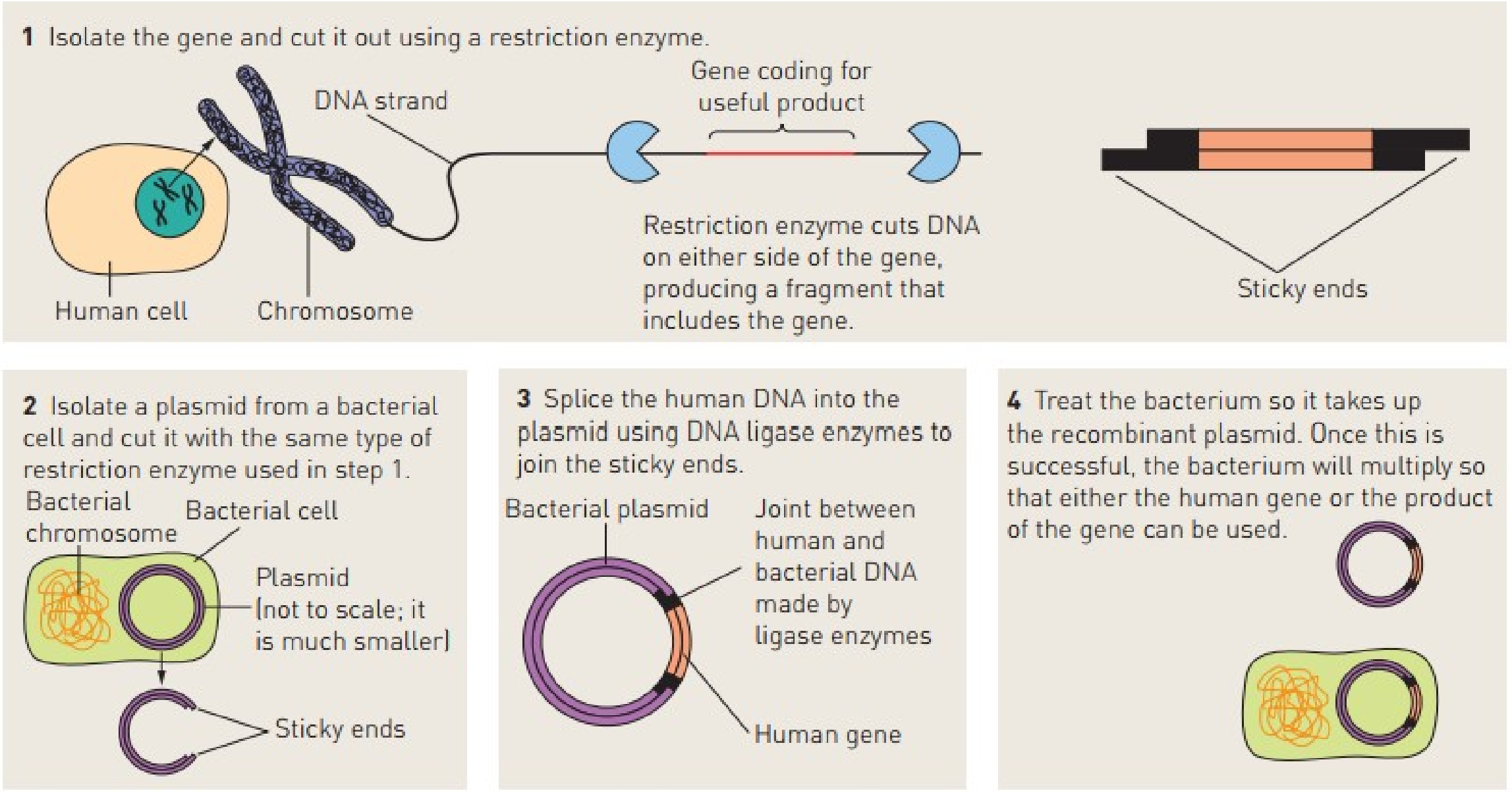
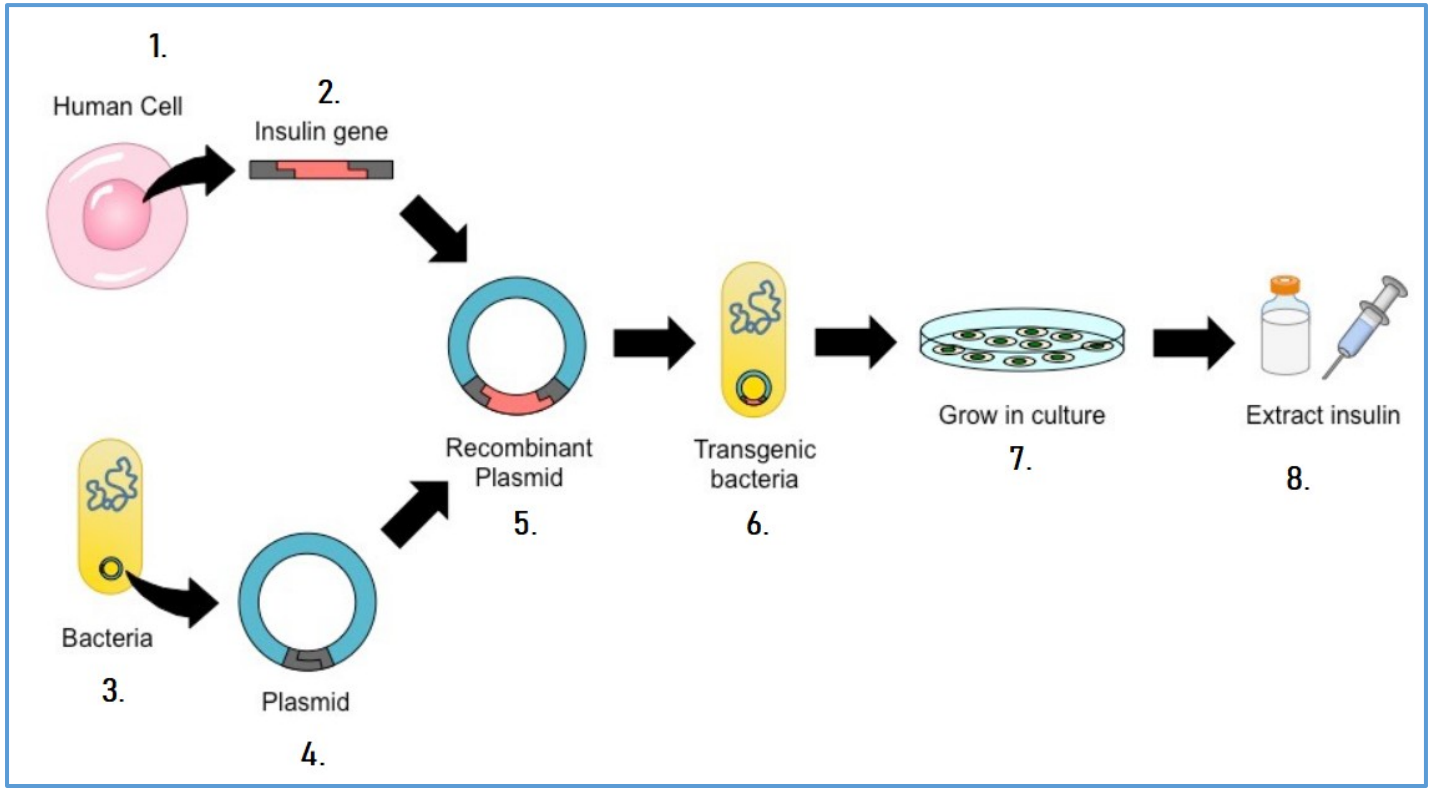


Figure 13.9 A simplified diagrammatic representation of recombinant DNA technology



Term	Definition
Blunt ends	The ends produced by a straight cut of a sequence of nucleotide bases
Ligase	An enzyme capable of combining two small components of single-strand DNA into one single structure
Phage	Or bacteriophage; a virus that infects bacteria
Plasmid	Small circular strand of DNA distinct from the main bacterial genome; it is composed of only a few genes and is able to replicate independently within a cell
Restriction enzyme	Enzyme that cuts a strand of DNA at a specific sequence of nucleotides
Staggered cut	Produced when a restriction enzyme creates fragments of DNA with unpaired nucleotides that overhang at the break in the strands; called sticky ends
Straight cut	Produced when a restriction enzyme makes a clean break across the two strands of DNA so that the ends terminate in a base pair; called blunt ends
Sticky ends	The overhanging ends produced by a staggered cut of a sequence of nucleotide bases; can be called cohesive ends
Vector	A bacterial plasmid, viral phage or other such agent used to transfer genetic material from one cell to another

1: The base sequence coding for insulin production (the insulin gene) is isolated from a healthy human cell.

2: The base sequence on either side of the gene is treated with a *restriction enzyme* that cuts the base sequence, leaving a *staggered cut* with *sticky ends*.

3: A *plasmid* is isolated from a bacterial cell.

4: The plasmid is treated with the *same* restriction enzyme that was used on the insulin gene so that it has the *same* staggered cut and the *same* sticky ends.

5: The human insulin gene is then inserted into the bacterial plasmid, which fits because the plasmid and the gene have the same staggered cut and sticky ends. The plasmid and insulin gene are stuck together using the enzyme *DNA Ligase*.

6: The recombinant plasmid acts as a *vector* (carrier) of the human insulin gene. It is inserted into a new bacterium, creating a *transgenic* bacterium that contains the gene for human insulin. Because it contains the gene for human insulin, the bacterium can now produce human insulin hormone.

7: The transgenic bacterium is grown in culture, where it reproduces until there is a large number of bacteria, all producing the human insulin hormone.

8: The bacterial culture is processed to extract the hormone insulin that the bacteria have produced and it is refined for human use, to treat Type I diabetes.

Examples of use

- Insulin
  - Pre-1982, diabetics treated with insulin from pig/cattle pancreas
  - 1982: insulin produced from genetically engineered bacteria approved – “Humulin”
  - Human gene spliced into bacterial cells, so bacteria would then produce insulin
  - Insulin now produced in huge vats of bacteria.

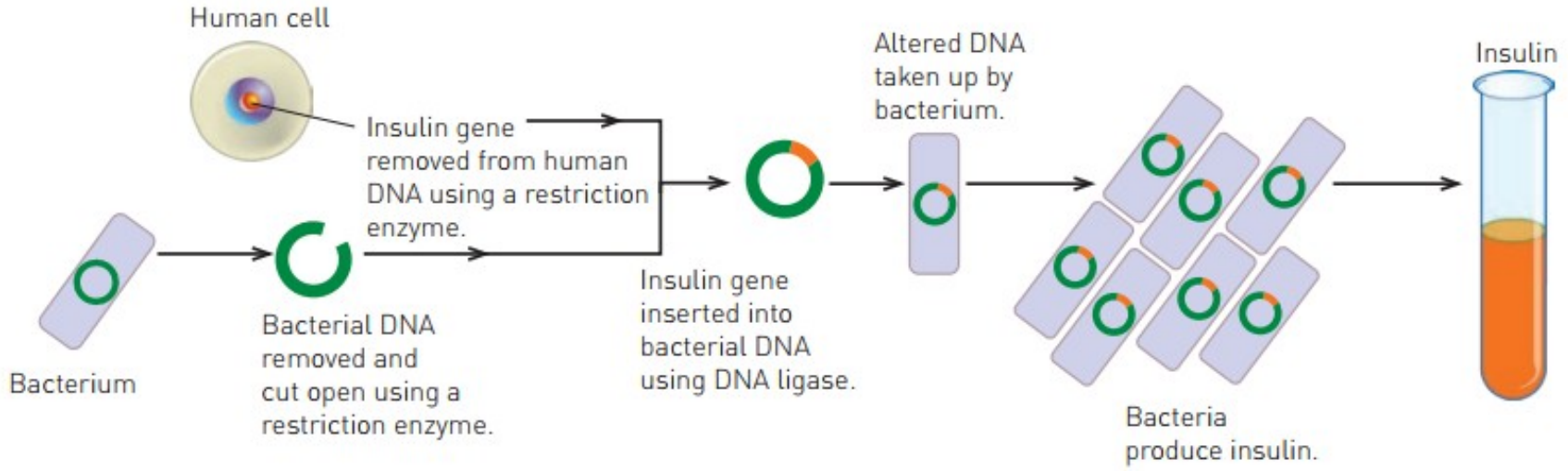


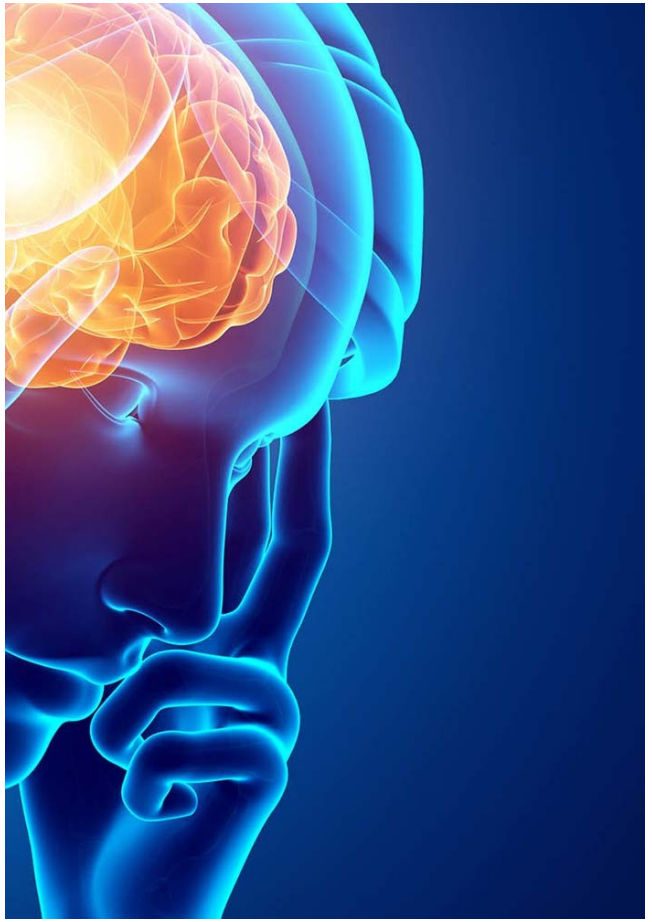
Figure 13.10 A diagrammatic representation of the production of human insulin using recombinant DNA technology

Examples of use continued

- Human Growth Hormone
  - Now produced by genetically engineered *E. coli* bacteria
  - Used to treat lack of growth hormone in pituitary disorders
  - Also used in dairy cattle to increase milk production

Note: the process for making synthetic Human Growth Hormone using Recombinant DNA technology is exactly the same as for making synthetic insulin, except the gene for Growth Hormone is isolated and worked with, not the gene for insulin.

Review: Diseases of the Nervous System and Biotechnology

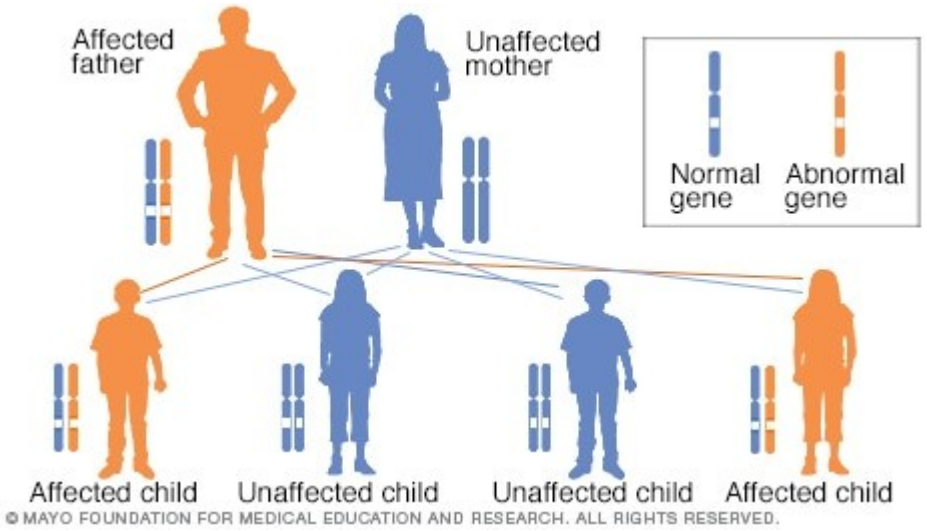


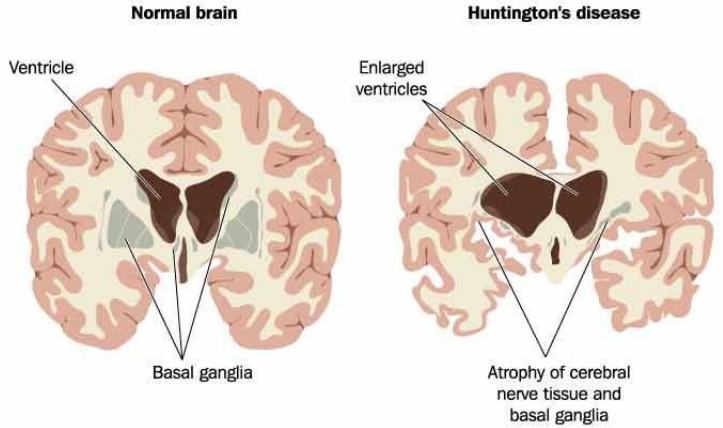
The Nervous System is highly organised

Technology is used to treat diseases

Huntington's Disease

- A single gene disorder – dominant inheritance.
- Abnormal gene:
  - Codes for a mutated protein (huntingtin)
  - Protein damages nerve cells in the brain
  - Large
- Symptoms start after age 40 and are progressive:
  - Physical, mental and emotional changes
  - Progressive dementia, limb flailing.
- Currently incurable, fatal.
- Possible future treatment: Gene therapy

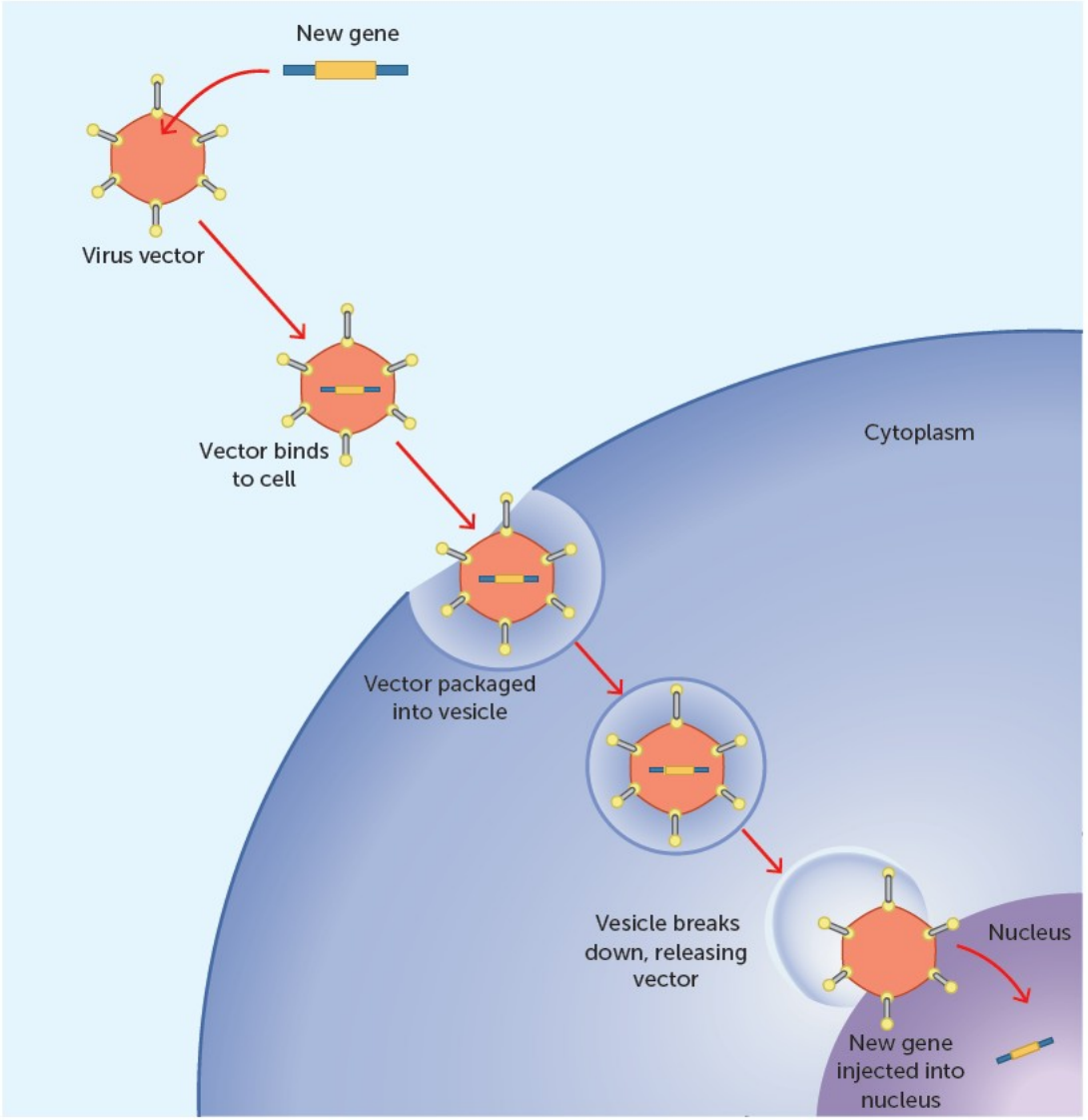




Gene Therapy

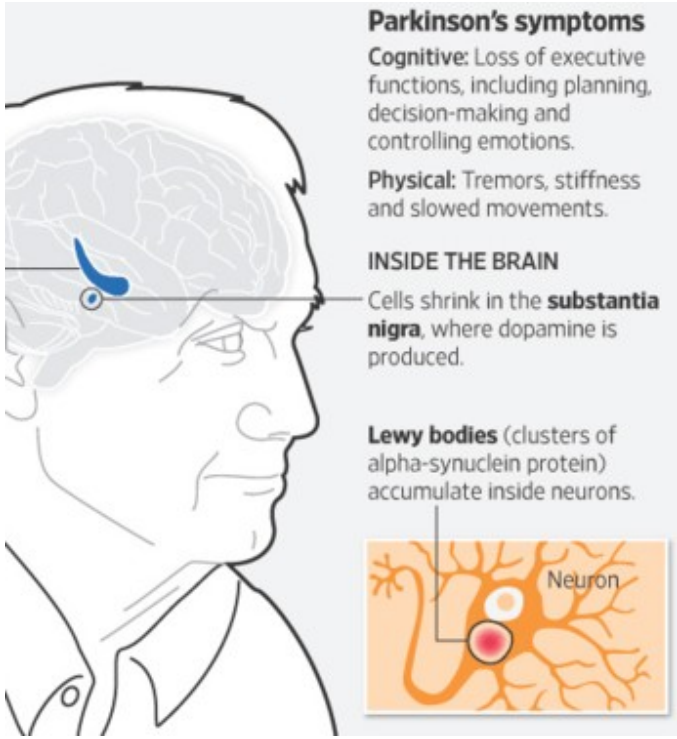
- Aims to treat or cure disease by identifying faulty genes and inserting healthy ones.
- Still in research phase.
- Concept is that a vector (eg a virus)can be used to deliver the desired DNA into the cell, to be incorporated by the nucleus and produce desired protein.
- Areas of possibility:
  - Replacing mutated gene with healthy copy*
    - Eg **Huntington's disease**, Cystic Fibrosis, Type I Diabetes
  - Fixing or inactivating mutant genes*
    - Eg **Huntington's disease**
  - Inserting a new gene that will fight the disease
  - Making the immune system recognise diseased cells.
    - Eg therapy for some cancers

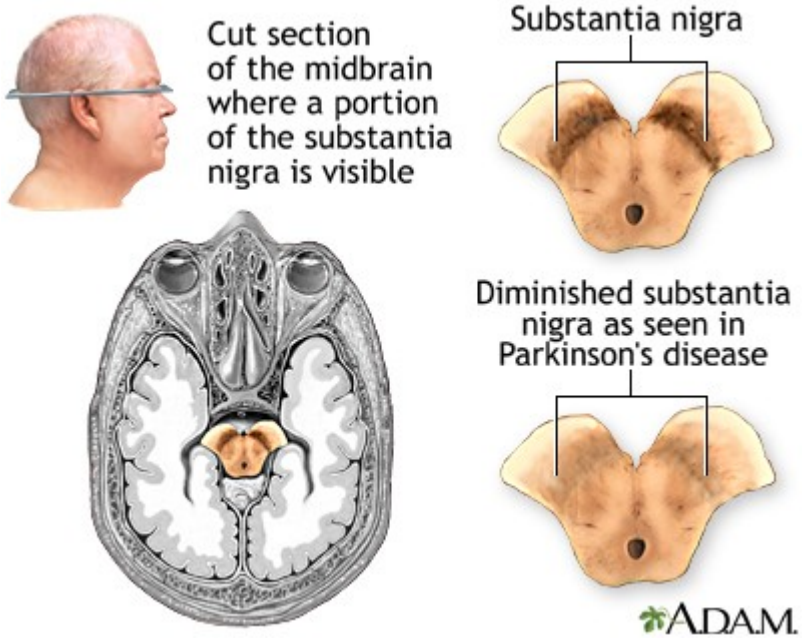
*These ones could be used for Huntington's Disease



Parkinson's Disease

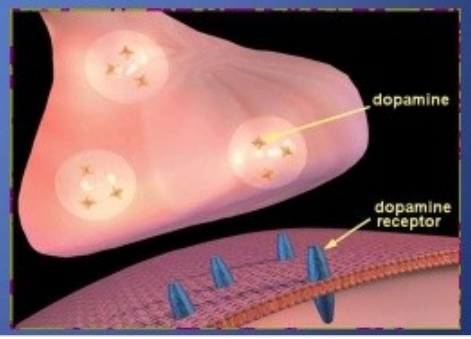
- Degenerative (progressive) neurological disease
  - Central NS tissues affected
    - Cell death in substantia nigra (in basal ganglia) decreases dopamine production causing symptoms
    - No identified cause of the cell death
    - Lewy bodies (protein) accumulates inside neurons
  - Symptoms: Impaired motor skills and speech
  - Usually diagnosed later in life (between 60 and 70 years old)
  - No cure, but can be managed.
  - Future cure may be cell replacement therapy





What is dopamine?

- Dopamine is a neurotransmitter in the brain
- Essential component of the basal ganglia motor loop as well as the neurotransmitter responsible for controlling the exchange of information from one brain area to another
- Plays vital roles in certain behavior such as
  - Movement
  - Cognition
  - Pleasure
  - Motivation



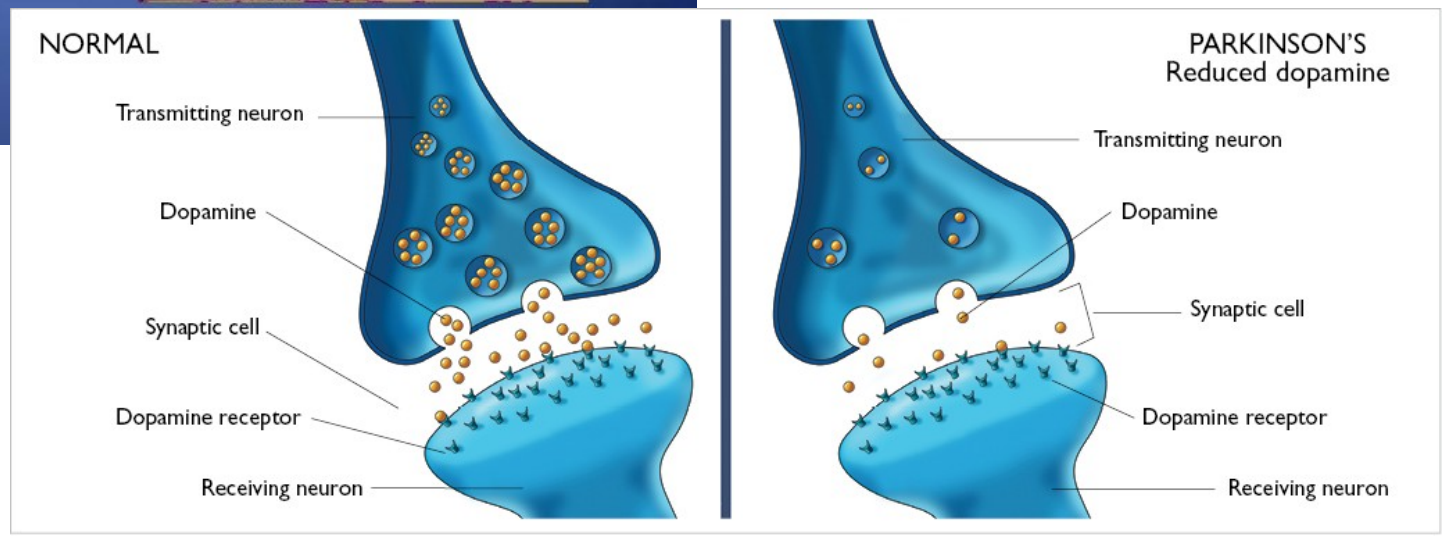
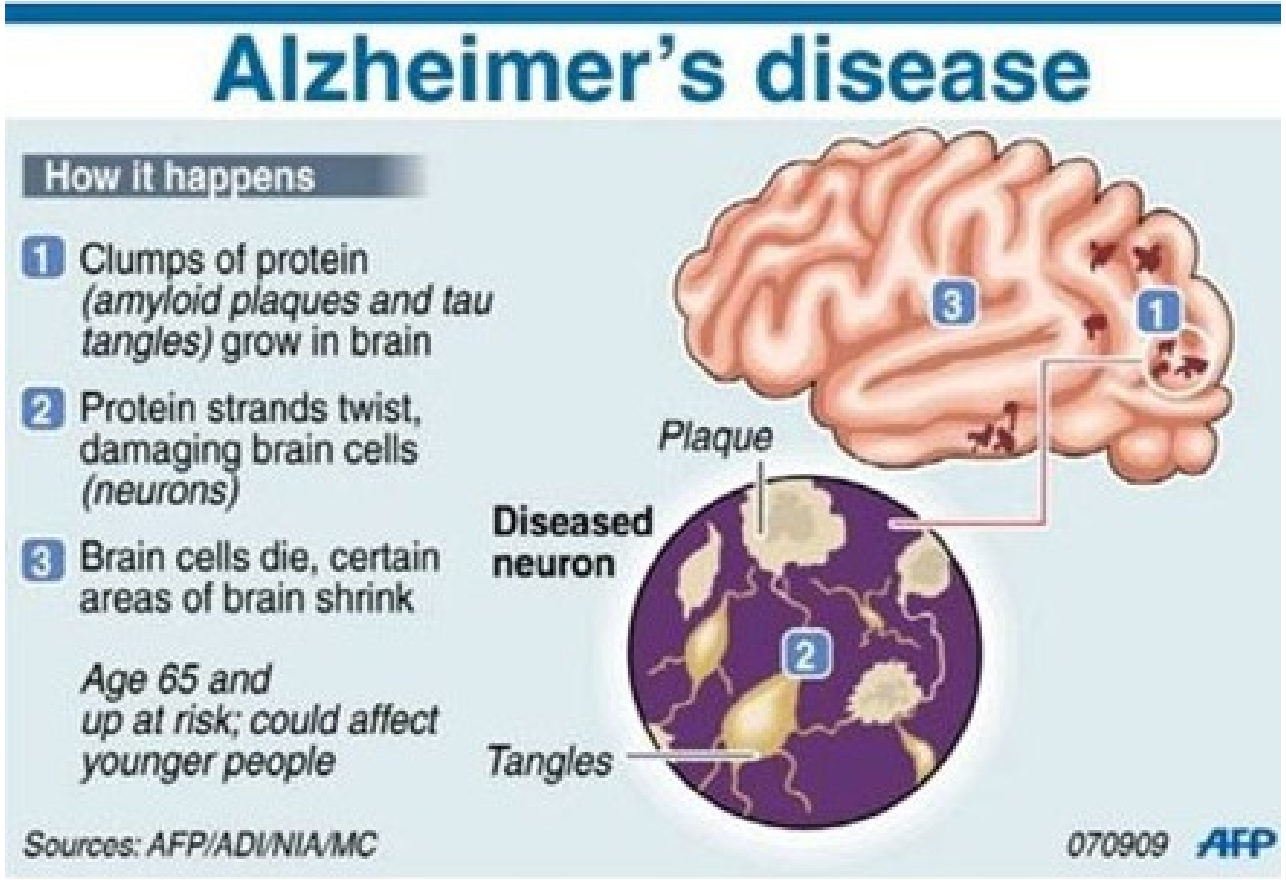


FIGURE 3. Illustration showing low levels of dopamine in a neuron affected by Parkinson's disease (right) and normal levels (left).

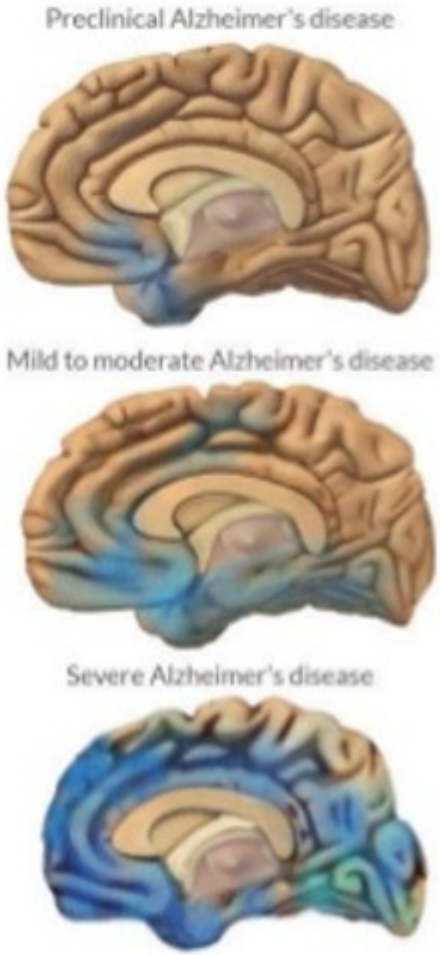
Alzheimer’s Disease

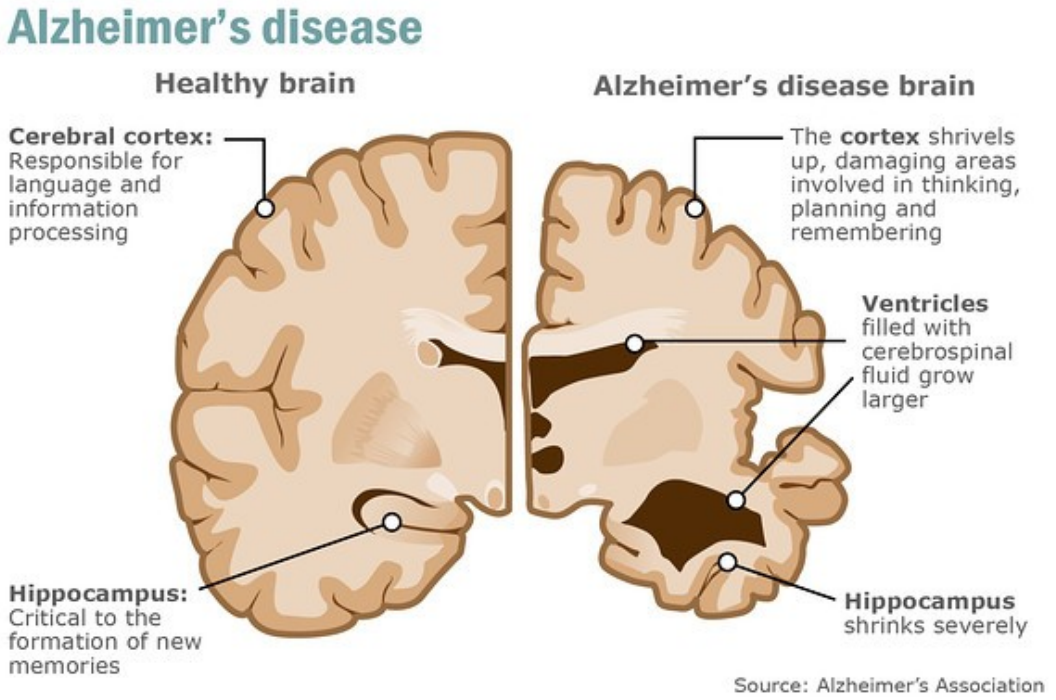
- Progressive Degenerative disease of the brain
- Usually diagnosed after the age of 65
- No clear cause has been established
- Disease progression:



As the disease progresses…

- Neurons lose their ability to function and communicate with each other, and eventually die.
- As it progresses, neurofibrillary tangles spread throughout the brain (shown in blue). Plaques also spread throughout the brain.
- By the final stage, damage is widespread and brain tissue has shrunk significantly.
- Currently no cure – future cure could be cell replacement therapy







Symptoms of Alzheimer's disease

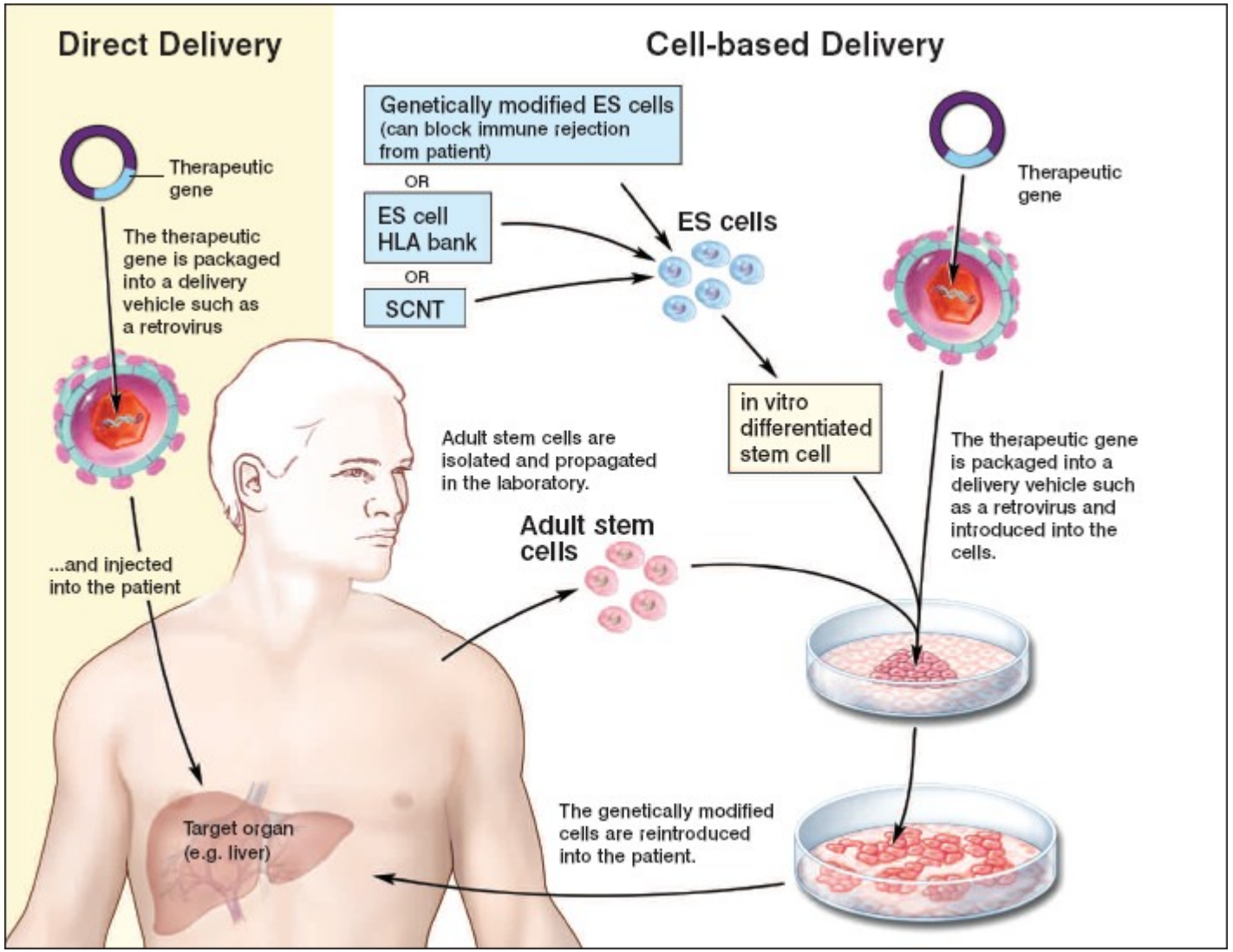
Cell Replacement Therapy

- Could be used in future for neurodegenerative disorders such as Alzheimer's and Parkinson's.

- Replacement of dying neurons with healthy ones derived from stem cells
  - Can grow and differentiate to replace damaged tissue
  - Controversy surrounding use of human embryonic stem cells
  - Other sources of stem cells are being researched

- Technique not yet refined – is a potential future treatment.
  - Adult stem cells are isolated from the patient.
  - These are treated so that they differentiate into the required tissue, and cultured in the laboratory.
  - The newly differentiated cells are inserted into the damaged areas to regrow healthy tissue.

How Cell Replacement therapy might work once it is through the experimental stage:



This example shows the liver as the target organ, but the idea is the same for diseases such as Alzheimer's and Parkinson's where a part of the brain is the target.